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Linear Algebra
Semester: Spring 2025

MATH 205: LINEAR ALGEBRA

SPRING 2025

Definitions and Formula Sheet

- A system of n equations with n variables:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = 0 \end{cases}$$

- Augmented matrix for the system $AX = B$:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{array}\right]$$

- If A is $m \times r$ and B is $r \times n$, then:

$$AB = \text{Matrix of size } m \times n.$$

- **Transpose of a Matrix:** For a 3×2 matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}, \quad A^T = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix}.$$

- A matrix is symmetric if $A^T = A$.
- A matrix is skew-symmetric if $A^T = -A$.
- **Elementary Row Operations:**
  1. Multiply a row by a nonzero constant: $(R_i \to cR_i,\ c \neq 0)$
  2. Interchange two rows: $(R_i \leftrightarrow R_j)$
  3. Add a multiple of one row to another: $(R_i \to R_i + cR_j)$
- **Finding A^{-1} using Row Operations:**

$$[A \mid I] \xrightarrow{E.R.O.S} [I \mid A^{-1}].$$

- If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $\det(A) = ad - bc$.
- For two square matrices A and B: $\det(AB) = \det(A)\det(B)$.
- For the identity matrix I_n: $\det(I_n) = 1$.
- For an invertible matrix A: $\det(A^{-1}) = \frac{1}{\det(A)}$.
- $\det(A) = \det(A^T)$.
- The determinant of a matrix A can be evaluated using cofactor expansion along any row or column:

$$\det(A) = \sum_j a_{ij}C_{ij},$$

  where C_{ij} is the cofactor of a_{ij}.
- The adjoint matrix $\text{adj}(A)$ is the transpose of the cofactor matrix and inverse is given by:

$$A^{-1} = \frac{\text{adj}(A)}{\det(A)}.$$

- **Cramer's Rule:** For a system $AX = B$, the i-th variable is:

$$x_i = \frac{\det(A_i)}{\det(A)},$$

  where A_i is obtained by replacing the i-th column of A with B.
- For an upper or lower triangular matrix, the determinant is the product of diagonal elements:

$$\det(T) = t_{11} \cdot t_{22} \cdots t_{nn}.$$

- An n-tuple is an element of $\mathbb{R}^n$. Vector addition and scalar multiplication in $\mathbb{R}^n$ follow:

$$\mathbf{u}+\mathbf{v} = (u_1+v_1, \ldots, u_n+v_n), \quad k\mathbf{u} = (ku_1, \ldots, ku_n).$$

- **Euclidean Inner Product:** For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i.$$

- **Norm and Distance:** The norm (length) and distance are defined as:

$$\|\mathbf{u}\| = \sqrt{\mathbf{u} \cdot \mathbf{u}}, \quad d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|.$$

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- **Cauchy–Schwarz Inequality:** For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:
$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\|\|\mathbf{v}\|.$$
- **Triangle Inequality:** For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:
$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$
- Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are orthogonal if: $\mathbf{u} \cdot \mathbf{v} = 0$. And for orthogonal vectors $\mathbf{u}$ and $\mathbf{v}$:
$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2.$$
- **Vector space axioms:**
  1. For all $\mathbf{u}, \mathbf{v} \in V$, $\mathbf{u} + \mathbf{v} \in V$.
  2. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ for all $\mathbf{u}, \mathbf{v} \in V$.
  3. $\mathbf{u}+(\mathbf{v}+\mathbf{w}) = (\mathbf{u}+\mathbf{v})+\mathbf{w}$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$.
  4. There exist $\mathbf{0} \in V$ such that $\mathbf{0}+\mathbf{u} = \mathbf{u}+\mathbf{0} = \mathbf{u}$ for all $\mathbf{u} \in V$.
  5. For all $\mathbf{u} \in V$ there exist $-\mathbf{u} \in V$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0} = -\mathbf{u} + \mathbf{u}$
  6. $k\mathbf{u} \in V$ for all $\mathbf{u} \in V$ and scalar k.
  7. $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in V$ and scalar k.
  8. $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$ for all $\mathbf{u} \in V$ and scalars k and m.
  9. $k(m\mathbf{u}) = (km)(\mathbf{u})$ for all $\mathbf{u} \in V$ and scalars k and m.
  10. $1\mathbf{u} = \mathbf{u}$ for all $\mathbf{u} \in V$.
- **Coordinate Vector:** For $\mathbf{v} = \alpha_1\mathbf{b_1} + \cdots + \alpha_n\mathbf{b_n}$, the coordinate vector is $[v]_B = \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}$.
- **Rank-Nullity Theorem:** $\text{rank}(A) + \text{nullity}(A) = n$, where A is an $m \times n$ matrix.
- **Norm and Distance:** $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$, $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.
- **Orthogonal Vectors:** $\mathbf{u} \cdot \mathbf{v} = 0$ for orthogonal vectors.
- **Gram-Schmidt:** $\mathbf{u_i} = \mathbf{v_i} - \sum \text{proj}_{\mathbf{u_j}}\mathbf{v_i}$, for orthogonalizing vectors.
- **Characteristic Polynomial:** $p_A(\lambda) = \det(A - \lambda I)$, where A is a matrix and λ is an eigenvalue.
- **Linear Transformation:** $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$, $T(k\mathbf{u}) = kT(\mathbf{u})$.
- **Kernel and Range:** $\ker(T) = \{\mathbf{v} : T(\mathbf{v}) = 0\}$, $\text{range}(T) = \{T(\mathbf{v}) : \mathbf{v} \in V\}$.